

SEUNGMIN HAN

Systems Architect · Capital Markets · AI-Augmented Engineering

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PROFESSIONAL EXPERIENCE

Korea Investment Securities (한국투자증권) — Investment Engineering Dept. · Seoul

Sep 2024 – Present

Market-Making / Liquidity-Provision (MM/LP) Systems Developer

- **Develop and own** the firm's MM/LP trading platform end to end — front-end UI (C# / DevExpress WinForms), back-end services (C# and C++), data & analytics, and hardware / infrastructure — quoting **~5,500 instruments** simultaneously on the Korea Exchange (KRX).
- **Re-architected** a 13-service .NET trading monolith into independently deployable services with a strangler-fig migration, cutting over risk-critical order logic safely by running the legacy engine as a parity oracle (no big-bang cutover).
- **Built** a C++20 ultra-low-latency order/hedge engine from scratch — core-pinned threads, zero-copy parsing, lock-free quote tables — targeting single-digit-microsecond tick-to-trade, and the foundation for migrating the firm's **Windows + C# platform to Linux + C++**.
- **Designed and deployed** a live KOSPI 200 index-arbitrage engine with fail-closed pre-trade risk gates, automated re-pricing, and full order-blotter reconciliation.
- **Optimized** the FEP exchange-gateway relay with the infrastructure team — tuned the TCP stack and applied **kernel-bypass** to spread orders — reaching **~50 μs** quoting latency (best-case ~25 μs).
- **Designed** the API and ledger-to-FEP synchronization algorithm for the firm's DMA direct-ledger gateway, letting the OMS send orders without delay while the gateway validates balance on every order.
- **Partnered** with the Information Security, IT Strategy, and Infrastructure divisions (정보보호부 · IT전략부 · 인프라부) to harden infrastructure and run a regular hardware-refresh cycle for uninterrupted trading uptime.

Supergate Inc. · Seoul

Summers 2022 & 2023

Software Developer / Research Intern — Applied Cryptography

- Researched and engineered a searchable database over **fully homomorphic encryption (FHE)** — arbitrary computation on encrypted data — in C++, supporting private exact-match and range queries without decryption.
- Co-invented two private-query protocols and filed **two Korean patent applications** ("Encrypted Reverse Index" and "PDQ Protocol over Finite Extension Fields"); the protocols cut query-result size **~50% on average**.
- Cross-compiled the database server and dependencies onto a **Xilinx Zynq FPGA** board for hardware-accelerated encrypted search.

Daangn Market (Karrot) · Seoul

2021

Global Business Development & Product Intern

- Drove international growth of the "Karrot" marketplace in Toronto, helping grow monthly active users **~1.5x**.
- Built a live in-app behavioral analytics dashboard (BigQuery + Looker/Data Studio) used to steer product decisions.
- Ran product-market-fit research and competitive analysis (design, retention, go-to-market) to shape a market-penetration strategy.

University of Pennsylvania · Philadelphia, PA

2021 – 2023

Head Teaching Assistant — Operating Systems (CIS 3800 / MCIT 5950)

- Led and coordinated the teaching-assistant team for 250+ students; ran recitations, code reviews, and office hours.
- Automated edge-case discovery in the homework auto-grader (Python), improving grading reliability.

Unbounded Research in Biomedical Systems Lab · Philadelphia, PA

2019

Machine-Learning Research Assistant

- Built an ensemble of Relief-based feature-selection algorithms (scikit-learn) achieving a **~1.5x speedup** on large biomedical datasets.

SELECTED PROJECTS

Self-Hosted Multi-Asset Portfolio & Finance Platform — Python/FastAPI · HTMX · TimescaleDB · DuckDB

- Real-time personal portfolio and finance platform spanning Korean equities, US equities/FX, and **crypto**, with live feeds (KIS OpenAPI, Binance futures, yfinance), server-sent-event push, PnL, and net-worth history.
- Engineered a stateless, **computed-from-source** debt model so monthly card debt clears itself with zero manual entry and cannot drift — a single-source-of-truth design.

AI-Augmented Engineering Harness — applied LLM / multi-agent systems

- How I now build large systems: a multi-agent harness I **direct to design, implement, and review** code — parallel-agent review with adversarial refutation and an independent second-model cross-check, end-to-end feature pipelines, and a self-maintaining decision / knowledge base. Engineering judgment, applied at AI throughput.
- Runs on a self-hosted compute mesh (Tailscale) — Mac orchestrator, always-on mini, wake-on-LAN Linux server — with git-backed sync and scheduled jobs.

Fully Secure Online Voting Platform using FHE — C++ · HElib · Senior Design, "Most Technically Challenging" award

- Led an end-to-end encrypted election system in which ballots are tallied entirely under encryption.

EDUCATION

University of Pennsylvania — School of Engineering & Applied Science · Philadelphia, PA

- **M.S.E., Computer & Information Science** — May 2024. Thesis: *Efficient Private Query Protocol using Fully Homomorphic Encryption*.
- **B.S.E., Computer & Information Science**, Minor in Mathematics — 2023. GPA 3.64/4.00; Dean's List (2021–2023); CIS Senior Design "Most Technically Challenging" award.

PATENTS (FILED)

- "Encrypted Reverse Index" — Korean patent application **10-2023-0115045**.
- "PDQ Protocol Using Finite Extension Fields" — Korean patent application **10-2023-0115051**.

EXPERTISE

Capital Markets & Trading Systems: market making / liquidity provision, index arbitrage, derivatives hedging, market data, exchange (KRX / FEP) connectivity, kernel-bypass networking, low-latency execution, OMS & ledger synchronization, pre-trade risk.

Cryptography & Security: fully homomorphic encryption, privacy-preserving protocols, applied security, crypto & market-data / PnL systems.

Engineering: C++20, C# / .NET 8, Python, TypeScript / React, SQL · Linux & Windows, DevExpress / WinForms · concurrency & lock-free systems, distributed & real-time architecture · FastAPI / HTMX, PostgreSQL / TimescaleDB · applied LLM / agentic systems.

Architecture, Product & Delivery: system architecture, AI-augmented delivery, product management, data-driven growth, technical-to-business translation, cross-functional leadership.

Languages: Korean (native), English (fluent — U.S. educated).